

Piezoelectric-actuated MEMS Dual-axis Micromirror

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Abstract—Micromirrors are essential components in MEMS-based optical systems, enabling dynamic light control for applications like displays and laser scanning. Traditional designs, often constrained to single-axis motion, limit their angular range and versatility. This work introduces a dual-axis piezoelectric-actuated micromirror system, featuring multiple piezoelectric bimorph beams arranged in series on all four sides of the micromirror. This configuration enhances displacement and angular range, enabling precise multi-axis control. The design overcomes challenges such as fabrication complexity and pull-instability seen in conventional comb-drive-based systems. This approach demonstrates potential improvements over conventional systems, addressing key limitations and advancing micromirror technology.

Index Terms—Micromirror, MEMS, Piezoelectric bimorph

I. INTRODUCTION

Micromirror systems, a fundamental component of MEMS-based optical devices, have revolutionized technologies like digital projection, optical switching, and maskless photolithography. By redirecting light through controlled mirror actuation, micromirrors enable dynamic light manipulation with unparalleled precision. The Digital Micromirror Device (DMD), widely used in projection displays, exemplifies their transformative impact.

However, conventional micromirror systems are predominantly constrained to single-axis motion, limiting their versatility in applications requiring multi-dimensional light redirection. Furthermore, traditional designs relying on comb-drive actuators face significant challenges, including:

- Fabrication complexity: Narrow comb-drive gaps demand high-resolution manufacturing.
- Spatial inefficiency: Comb drives occupy substantial surface area, restricting compact designs.
- Mechanical vulnerabilities: Pull-in instability and susceptibility to external noise compromise performance and reliability.

To overcome these limitations, we propose a novel dual-axis micromirror system actuated by piezoelectric bimorph beams. Each side of the micromirror incorporates four bimorph beams arranged in series, amplifying displacement and therefore angular range. These bimorphs, composed of piezoelectric layers with opposing poling directions, deform under applied voltage, enabling precise multi-axis motion.

In this paper, we detail the design, fabrication, and performance evaluation of the proposed system, highlighting its potential to transform applications in optical redirection and scanning.

II. DESIGN AND OPERATION

The piezoelectric-actuated dual-axis micromirror operates by utilizing four sets of bimorph piezoelectric beams. When a voltage is applied, each set of beams bends individually, generating distinct tilt angles of the mirror to direct light as needed. A COMSOL 3D model of the micromirror is shown in Figure 1, and a top schematic view is provided in Figure 3 (not to scale).

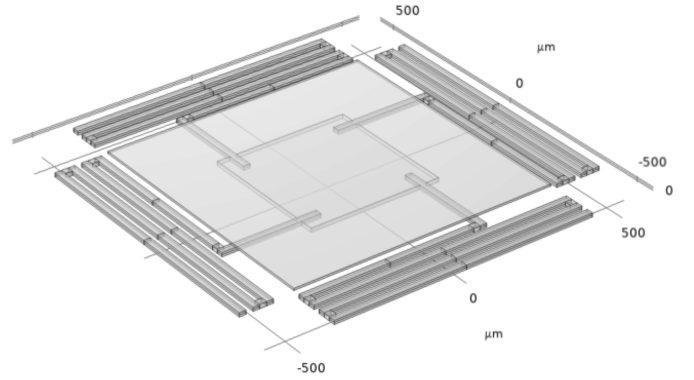


Fig. 1. COMSOL 3D model of the micromirror device

A. Design

The micromirror consists of a thin layer of high-reflectivity material, which serves as the mirror, coated above a square-shaped stage made of Si_3N_4 . The stage has a side length of 800 μm and a thickness of 8 μm . This stage is supported by a PZT-5H base, measuring 400 μm on each side. A bimorph beam extends from each side of the base, made of two 8 μm -thick layers of PZT-5H piezoelectric material sandwiched between three aluminum electrode layers (of negligible thickness). PZT-5H is selected for its high piezoelectric strain coefficient (d_{31}) and low Young's modulus (E), enabling greater deflection under applied voltage compared to other piezoelectric materials.

The beams extend from the base and connect to a set of four bimorph beams arranged in a meander flexure design. The four beams each have a width of $20\ \mu\text{m}$: the beam closest to the mirror stage measures $300\ \mu\text{m}$ in length, the second one measures $520\ \mu\text{m}$, and the other three beams are each $800\ \mu\text{m}$ long. This configuration allows for the accumulation of deflections from each individual beam, resulting in a larger overall bending on each side. The entire mirror stage and bimorph beam flexures are suspended above a silicon substrate with a Si_3N_4 isolation layer, with the outermost beam anchored to the substrate.

B. Operation

To actuate the micromirror, the aluminum electrode at the center is grounded, while positive or negative voltages are applied to the top and bottom electrodes. Each beam has two opposite voltage polarities distributed (as shown in the left beam flexure of Figure 3). This configuration simulates the guided-end effect, which helps minimize the tip angle (Figure 2). This is beneficial for reducing the stress accumulation at the connections between the beams.

As shown in Figure 3, assuming the orientation of the PZT-5H material is perpendicular to the page, the top PZT-5H layer at the upper section of the outermost beam contracts along its length, while the bottom layer extends. This results in the outermost beam deflecting out of the page. Following this logic, the deflections of the four beams accumulate, causing the left side of the mirror to be lifted.

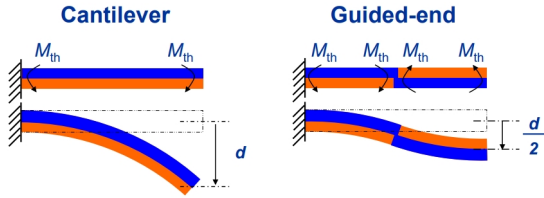


Fig. 2. Cantilever and Guided-end beams bimorph deflections [1]

To control the tilt angle of the micromirror, four separate analog power supplies are required to independently adjust the magnitude of bending in each of the bimorph beam flexures. This allows precise control over the deflection at each side of the mirror stage, thus enabling fine-tuning of the tilt angle.

III. FABRICATION

The device fabrication process is shown schematically in Figure 4, outlining the essential stages (with sections A-A and B-B shown in Figure 3). A key challenge lies in applying a high reflectivity coating, which is crucial for ensuring efficient light reflection. The flatness of the Si_3N_4 layer must be precisely controlled, as any surface irregularities could lead to performance degradation. Minimizing surface roughness is vital to reduce optical losses and ensure the precision required for optimal functionality.

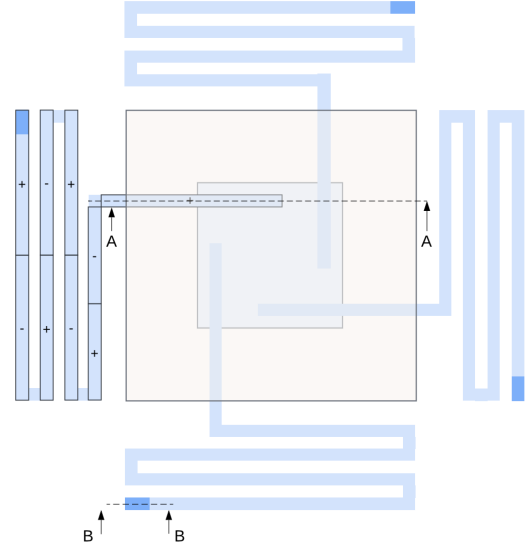


Fig. 3. Top schematic view of micromirror device

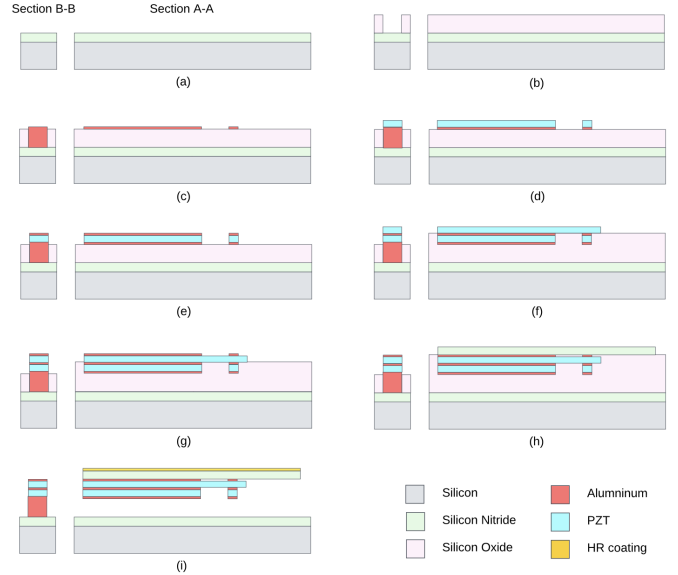


Fig. 4. Fabrication steps for the micromirror design. (a) The Silicon wafer is processed and a layer of Si_3N_4 is grown for isolation using LPCVD. (b) A sacrificial layer of SiO_2 is deposited and photolithography is used to define anchor patterns. (c) A layer of Al is deposited and etched to form the bottom electrodes, and also serve as mechanical anchors for the structure. (d) PZT-5H is deposited and etched to define the beam structures. (e) A second layer of Al is deposited and etched to create the middle electrodes. (f) A second sacrificial layer of SiO_2 is deposited and polished, followed by the deposition and etching of a second layer of PZT-5H for the beams and the base of the micromirror stage. (g) A final layer of Al is deposited for the top electrodes. (h) SiO_2 is deposited and polished, followed by the deposition and etching of Si_3N_4 for the micromirror stage. (i) A high-reflectivity coating is applied to the micromirror stage. The structure is then released using HF wet etching, which removes the sacrificial SiO_2 layer and frees the device for operation.

IV. EXPECTED RESULTS

A. Expected Angle of Operation

The tilt angle θ of the micromirror plate is calculated considering alternating polarity in the deflections of 4 beams in series on one side and a perpendicular beam beneath the mirror. The total deflection (z_{tot}) is the sum of the deflections from 3 outer beams (z_{outer}), the fourth beam ($z_{4\text{th}}$), and the perpendicular beam ($z_{\text{perpendicular}}$).

For a cantilever beam, the deflection is given by:

$$z_{\text{beam}} = \frac{1}{2} d_{31} V \frac{L_{\text{beam}}^2}{t_{\text{beam}}^2},$$

where d_{31} is the piezoelectric strain coefficient, V is the applied voltage, L_{beam} is the beam length, and t_{beam} is the beam thickness. As illustrated in Figure 2, dividing this deflection by half will give the deflection of the guided-end beams in the design.

The parameters used in these calculations are as follows:

- $d_{31} = -127.84 \text{ pm/V}$
- Voltage: $V = 20 \text{ V}$
- Beam thickness: $t_{\text{beam}} = 6 \mu\text{m}$
- Beam lengths:

$$L_{\text{outer}} = 800 \mu\text{m},$$

$$L_{4\text{th}} = 520 \mu\text{m},$$

$$L_{\text{perpendicular}} = 300 \mu\text{m}$$

- Mirror base length: $L_{\text{plate}} = 400 \mu\text{m}$

These values allow us to calculate the deflections for each guided-end beam:

$$z_{\text{outer}} \approx -11.36 \mu\text{m},$$

$$z_{4\text{th}} \approx -4.80 \mu\text{m},$$

$$z_{\text{perpendicular}} \approx -1.60 \mu\text{m}$$

Using the fact that the beams on each side are in series, the total deflection is:

$$z_{\text{tot}} = 3 \cdot z_{\text{outer}} + z_{4\text{th}} + z_{\text{perpendicular}},$$

which gives:

$$z_{\text{tot}} \approx -40.48 \mu\text{m}$$

Considering the geometry, where one beam series lifts one side of the mirror while the other brings it down, the tilt angle θ is calculated as:

$$\theta = \arctan\left(\frac{z_{\text{tot}}}{L_{\text{plate}}/2}\right),$$

which results in:

$$\theta \approx -0.20 \text{ rad} \approx -11.5^\circ.$$

B. Stresses and Failure

The failure and stresses of the PZT beams and micromirror are analyzed based on the voltage applied, the thickness of the beams, and material properties of PZT-5H. The mechanical properties of PZT, including fracture strength (100 MPa) [4] and depolarization stress (30 MPa), are considered.

The stress that allows the beams to contract and expand is calculated using the formula:

$$\sigma = d_{31} E_3 E = d_{31} \frac{V}{t} E,$$

where $E \approx 60 \text{ GPa}$ is the Young's modulus of PZT-5H. [5] The resulting stress in the beams is thus:

$$\sigma \approx 25.6 \text{ MPa}.$$

This result indicates that the stresses in the beams are below the fracture strength of PZT-5H(100 MPa). However, they approach the depolarization threshold (30 MPa). The beams are structurally safe but may risk depolarization under prolonged operation or higher voltages.

V. SIMULATED RESULTS AND DISCUSSION

Based on the COMSOL simulation result, when opposite voltage polarities are applied on opposite sides of the mirror, the maximum tilt angle achieved is approximately $\pm 8.5^\circ$ with a voltage of $\pm 20 \text{ V}$ applied to the bimorph beam flexures (as shown in Figure 5). This results in a total light cone of approximately 17° . It is important to note that the tilt angle can be varied depending on which side of the mirror the voltage is applied. For example, to achieve a maximal diagonal tilt, the same voltage can be applied to two consecutive beam flexures to lift one corner of the mirror, while opposite polarities applied to the other two beam flexures will pull the opposite corner down, resulting in a diagonal tilt. This method offers versatile control over the micromirror's tilt in multiple directions.

The size of the micromirror alone (excluding the peripheral circuit) is $1100 \mu\text{m}$, which is a typical size for micromirrors used for laser scanning and wave guiding. Comparing its performance and size with some commercialized dual-axis micromirror devices based on comb drives, our design achieves a similar tilt angle but within a smaller size (S7644 Gimbal-Less Two-Axis Scanning MEMS Micromirrors from Mirrorcle Technologies, Inc. [2], with maximum tilt angle $\pm 6.1^\circ$, die size $4250 \mu\text{m}$ and mirror diameter $800 \mu\text{m}$; S13124-01 Electromagnetically driven two-dimensional laser scanning MEMS mirror from Hamamatsu TV Co. [3], with maximum tilt angle $\pm 10^\circ$, die size 13.2 mm and mirror diameter 1.95 mm).

However, the challenges of high power consumption and the potential for fracture at the flexure connection under high loads still persist. One possible solution is to use piezoelectric materials with a higher piezoelectric strain coefficient d_{31} and a higher fracture point. These materials could provide the

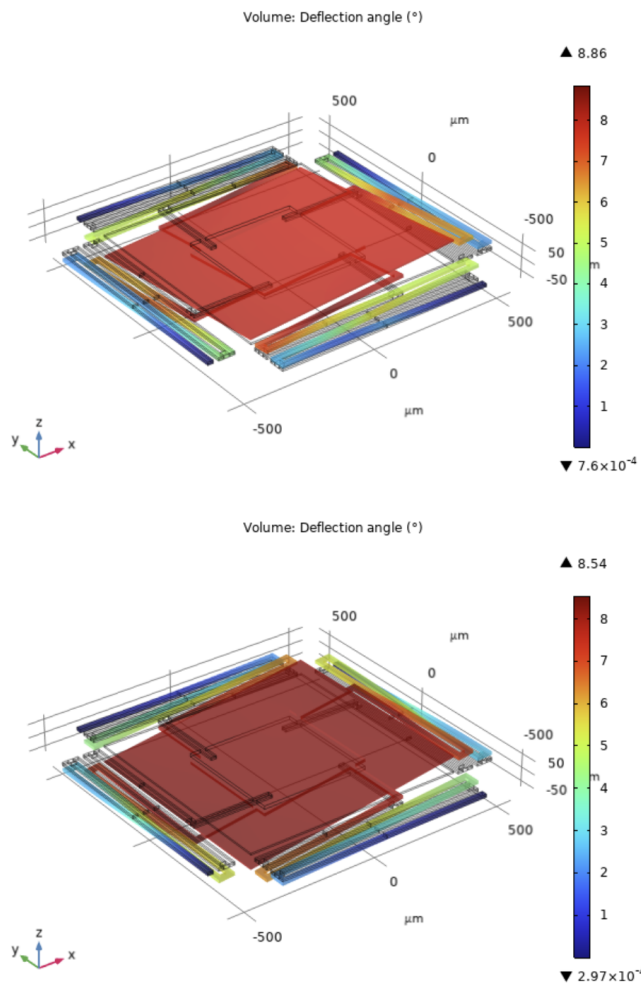


Fig. 5. Upper: Deflection of the micromirror along its diagonal, +20/0/-20V applied to two beam flexures and opposite polarity applied to another two; Lower: Deflection of the micromirror along its edge, +20/0/-20V applied to one beam flexure and opposite polarity applied to another, two other flexures idle

necessary deflection and durability, but the associated costs of these new materials and their compatibility with existing fabrication technologies need to be carefully evaluated to determine feasibility.

CONCLUSION

This work presents a novel piezoelectric-actuated dual-axis MEMS micromirror that overcomes the limitations of traditional single-axis optical MEMS designs. By utilizing four series-arranged bimorph piezoelectric beams, the proposed micromirror offers flexible tilt control, enabling complex multi-directional light redirection through strategic voltage manipulation.

The key innovation of this design lies in its ability to achieve intricate tilt configurations by selectively applying voltages across different beam flexures. However, future iterations should address the shear stresses at the interface between the

layers of the bimorph beams, as these stresses can influence the device's long-term reliability and performance. Despite these challenges, the proposed micromirror demonstrates a 17-degree light cone within a compact 1.1-millimeter device.

Future developments could focus on amplifying tilt angles by incorporating additional beams in series, allowing for a larger angular range without a proportional increase in material stress, potentially enhancing both performance and durability.

REFERENCES

- [1] G. Piazza, "18-614/24-614: Microelectromechanical Systems: Thermal MEMS," [PowerPoint Slide], pp.22, Fall 2024.
- [2] Mirrorcle Technologies, Inc., "Gimbal-Less Two-Axis Scanning MEMS Micromirrors," S7644 datasheet, https://www.mirrorcletech.com/pdf/DSI/MirrorcleTech-Datasheet_A318.2_800AL.pdf.
- [3] Hamamatsu TV Co., "Electromagnetically driven two-dimensional laser scanning MEMS mirror," S13124-01 datasheet, https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/ssd/s13124-01_koth1010e.pdf
- [4] A. J. Bell, T. P. Comyn, and T. J. Stevenson, "Expanding the application space for piezoelectric materials," APL Materials, vol. 9, no. 1, pp. 010901-1–010901-12, Jan. 2021, doi: 10.1063/5.0035416.
- [5] Piezo Systems, Inc. (n.d.). Piezoelectric material properties of PZT-5A and PZT-5H. Piezo Systems. Retrieved December 9, 2024, https://info.piezo.com/hubfs/Data-Sheets/piezo-PZT-5A_PZT-5H-material-properties.pdf